HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **advanced analytical models** on **Python** and **R** to support business process improvements for API manufacturing workflows, enhancing decision-making quality for process-centric teams.
- Developed and sustained a **data aggregation platform** using **Python** and **SQL** for process control and analysis, supporting continuous improvement initiatives across site departments.
- Provided **SME support** to local data viewers and consumers on **data analytics** tools and methodologies, facilitating wider adoption of data-driven insights.
- Collaborated with Process Centric Teams to understand technical business applications, applying **modelling techniques** to improve effectiveness and efficiency in API manufacture.
- Contributed to site-wide **data science business improvement projects**, identifying and implementing **Continuous Improvement (CI) opportunities** through data analysis.
- Delivered routine **statistical support** for investigations and continuous process verification (CPV) activities, ensuring adherence to **cGMP** standards for manufactured products.
- Containerized **Python**-based data analysis services using **Docker** and deployed them to an **AWS EKS** cluster, enabling scalable and reliable data processing for manufacturing intelligence.

SKILLS

Programming Languages – Python, R, SQL, Visual Basic

Data Science & Analytics – Data Modelling, Statistical Analysis, Machine Learning, Data Engineering, Data Aggregation, Predictive Modelling, Data Visualization

Tools & Platforms – Power BI, Tableau, scikit-learn, pandas, NumPy, TIBCO Statistica Informa, AWS, Docker, Kubernetes, Jenkins, Git

Methodologies – Agile, Scrum, Continuous Improvement (CI), Lean, Six Sigma, cGMP

Domain Knowledge – Pharmaceutical Manufacturing, API Processes, Investigations, Change Implementation, Process Control

PROJECTS

API Manufacturing Process Optimization Platform

- Developed a **Python**-based data aggregation platform to collect and analyze process parameters from manufacturing equipment, improving data visibility for **API production**.
- Implemented **statistical analysis** and **data modelling** using **scikit-learn** and **R** to identify key drivers of process variability, leading to a targeted **15% reduction** in batch deviations.
- Designed and built interactive **dashboards** in **Power BI** to visualize process performance and key metrics for site leadership and operational teams.

Predictive Maintenance Model for Manufacturing Equipment

- Engineered **Python** data pipelines to extract features from sensor data (temperature, pressure, vibration) for over 50 critical manufacturing assets.
- Trained a **gradient-boosted classification model** (XGBoost) on a dataset of ~300k historical readings to predict equipment failure, achieving an **AUC of 0.89** on held-out validation data.
- Deployed the trained model as a **FastAPI** service on **AWS Lambda**, providing real-time failure probability scores to the site's maintenance management system.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)